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$$f' = \frac{f}{\left(1+\frac{v_s}{v}\right)} \qquad (11.87)$$

Using binomial expansion again, we get,

$$f' = f\left(1-\frac{v_s}{v}\right) \qquad (11.88)$$

11.11.2 Observer in motion and source at rest

(a) Observer moves towards Source

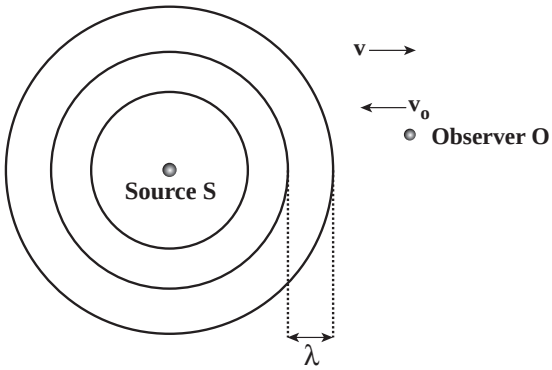


Figure 11.47 Observer moves towards Source

Let us assume that the observer O moves towards the source S with velocity v_o. The source S is at rest and the velocity of sound waves (with respect to the medium) produced by the source is v. From the Figure 11.47, we observe that both v_o and v are in opposite direction. Then, their relative velocity is $v_r = v + v_0$. The wavelength of the sound wave is $\lambda = \frac{v}{f}$, which means the frequency observed by the observer O is $f' = \frac{v_r}{\lambda}$. Then

$$f' = \frac{v_r}{\lambda} = \left(\frac{v+v_0}{v}\right)f = f\left(1+\frac{v_0}{v}\right) \qquad (11.89)$$

(b) Observer recedes away from the Source

If the observer O is moving away (receding away) from the source S, then velocity v_0 and v moves in the same direction. Therefore, their relative velocity is $v_r = v - v_0$. Hence, the frequency observed by the observer O is

$$f' = \frac{v_r}{\lambda} = \left(\frac{v-v_0}{v}\right)f = f\left(1-\frac{v_0}{v}\right) \qquad (11.90)$$

11.11.3 Both are in motion

(a) Source and observer approach each other

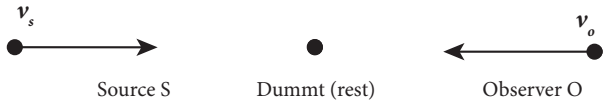


Figure 11.48 Source and Observer approach towards each other.

Let v_s and v_0 be the respective velocities of source and observer approaching each other as shown in Figure 11.48. In order to calculate the apparent frequency observed by the observer, as a simple calculation, let us have a dummy (behaving as observer or source) in between the source and observer. Since the dummy is at rest, the dummy (observer) observes the apparent frequency due to approaching source as given in equation (11.85) as

$$f_d = \frac{f}{\left(1-\frac{v_s}{v}\right)} \qquad (11.91)$$

At that instant of time, the true observer approaches the dummy from the other side. Since the source (true source) comes in a direction opposite to true observer, the dummy (source) is treated as stationary source for the true observer at that instant. Hence, apparent frequency when the true observer approaches the stationary source (dummy source), from equation (11.89) is